

Object Oriented Programming Lab

Lab 04

Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```

/*****
Lab:    xx
Task:   yy
Developer:
        Roll #: XXXYYYZ000
        Name:  XYZ
*****/

```

What you have to do

Program the following tasks in C++, compile and execute them.

Task

Create a class **Student** with:

- Private data members:
 - string name
 - int age
 - float gpa
- Public member functions:
 - void setName(string n)
 - void setAge(int a)
 - void setGPA(float g)
 - string getName() const
 - int getAge() const
 - float getGPA() const

(Use const with getter functions to ensure they do not modify the object.)

In main():

- Create **two student objects** and set their values using setters.
- Display their values using getters.
- Create a **pointer to a Student object**, set values using -> operator, and display details.

Home Task

Design a class called **Date**. The class should store a date in three integers: *month*, *day*, and *year*. There should be member functions to print the date in the form of **October 03, 2025**. Demonstrate the class by writing a complete program implementing it.

Input Validation:

Do not accept values for the day greater than 31 or less than 1.

Do not accept values for the month greater than 12 or less than 1.

😊😊😊 **BEST OF LUCK** 😊😊😊
